

AGENTX — AI Resume Intelligence & Autonomous Career Platform

Comprehensive Project Documentation & Technical Specification
Project Author & Architecture: Sandya Kaki & FastAPI Architecture | Version 1.0 Live

Resource	Production URL / Reference
Frontend Live URL	https://frontend-pi-lime-21.vercel.app
Backend Live URL	https://ai-resume-screener-26ke.onrender.com
GitHub Repository	https://github.com/sandya936/Ai_Resume_Screener
Pytest Harness	44 / 44 Passing Automated Unit Tests (100%)

1. Executive Summary & Planning

AGENTX is a production-oriented AI Resume Intelligence SaaS platform designed to transform static resumes into actionable, explainable career development plans. Built on a high-performance Modular Monolith architecture in FastAPI (Async Python 3.11) and Next.js 14+ (App Router), AGENTX replaces arbitrary black-box LLM scoring with an explainable **60% Deterministic + 40% AI Auditor Engine**, a Skill Normalizer, a 5-Category Open Mathematical Matching Formula, and an observable 6-Agent Autonomous System.

Development Planning Methodology:

- **Phase 1: Foundation & Domain Engine** — Pydantic domain models, magic-byte document validation, async ORM storage layer.
- **Phase 2: ATS Scoring & Skill Normalization** — 100+ Action verb registry, canonical keyword normalizer (K8s -> kubernetes), 5-category match score equation.
- **Phase 3: Multi-Agent Swarm Orchestration** — Built 6 specialized agents executing sandboxed tools with execution trace logs.
- **Phase 4: Dual-Theme Frontend UI** — Next.js 14 App Router UI with Warm Crimson Parchment vs Dark Cyber Slate theme switcher, guest demo login, and 40-question STAR interview generator.
- **Phase 5: Cloud Deployment & Testing** — 44/44 passing automated pytest suite, Vercel frontend deployment, Render backend deployment.

2. Technical Stack Specification

Layer	Technology	Purpose / Details
Backend Core	FastAPI (Python 3.11+)	Async ASGI Web Framework for high concurrency
Frontend Core	Next.js 14 App Router	React Framework with TypeScript & Server Components
Styling & CSS	Vanilla CSS3 & Tokens	Dynamic CSS variables, glassmorphism, responsive grids
Database	SQLAlchemy 2.0 Async	SQLite / PostgreSQL with Alembic migration support
AI Providers	Gemini 2.0 / OpenAI	LLM Provider Abstraction (BaseLLMProvider) & fallback
Security	PII Redactor & Headers	PII Log Redaction Filter, HSTS, CSP, X-Request-ID

Test Suite	Pytest Asyncio	44 Automated Tests covering APIs, models, agents
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3. System Architecture & Autonomous Multi-Agent Swarm

The system uses a modular monolith backend architecture communicating with a 6-Agent Manager-Worker Swarm System executing sandboxed tools:

- **ManagerOrchestratorAgent:** Plan decomposition, state aggregation, and trace graph management.
- **ResumeAgent:** Executes `tool_fetch_resume` for section extraction and candidate profile building.
- **JobDescriptionAgent:** Executes `tool_parse_job` for requirements, preferred skills, and experience extraction.
- **SkillGapAgent:** Executes `tool_evaluate_match` using canonical skill normalization and 5-category scoring.
- **RecommendationAgent:** Executes `tool_generate_roadmap` for prioritized 30-day learning curriculum.
- **InterviewAgent:** Executes `tool_generate_interview_questions` for 40-question STAR practice generator.

4. UI Design & Aesthetic Architecture

AGENTX features a dual-theme switcher (persisted via `localStorage` key `agentx_theme`):

1. **Warm Crimson Parchment Theme:** Background `#F4EFEB`, Crimson Accent `#8B0000`, Warm Sandstone Cards, Dark Espresso Typography `#2C221E`.
2. **Dark Cyber Slate Theme:** Background `#0B0F17`, Electric Emerald Accent `#10B981`, Obsidian Cards, Crisp Slate Typography `#F9FAFB`.

5. Implementation Features & Mathematical Matching

5-Category Weighted Match Formula:

Match Score = (Required Skills × 0.40) + (Preferred Skills × 0.15) + (Experience × 0.25) + (Projects × 0.10) + (Education × 0.10)

40-Question STAR Interview Practice Studio:

Generates 10 Technical Deep Dives, 10 Behavioral STAR Scenarios, 8 Project Experience Dives, 6 HR & Fit Questions, and 6 Job Description Specific questions with real-time answer scoring.

6. Verification Test Suite & Cloud Deployment

Automated Test Suite Results: 44 / 44 PASSED (100%) across 12 test modules including ATS analysis, authentication, career intelligence, job matching, multi-agent execution, and PII security headers.

Cloud Deployment Topology:

- **Backend:** Deployed on Render (Python 3.11 ASGI Uvicorn Server).
- **Frontend:** Deployed on Vercel (Next.js 14 App Router with edge optimization).